Media Formats of Advertising

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Abstract

Advertising is often disseminated through media. The evolution of media technologies has reshaped the advertising landscape over time. We present a model to shed light on this evolution, elucidating the important role of media format in advertising. The model features media's steering of consumer attention over time. Consumers, in turn, can decide where to direct their attention, given the media's allocation policy of advertising and media content. Media technologies can constrain the allocation capability, resulting in three fundamentally different advertising formats: static, sequential, and interactive. We illustrate how different formats grapple with two fundamental problems in media advertising: incentive misalignment between consumers and media, and media's inability to observe how consumers process ads. By identifying the conditions that distinguish and link different advertising formats, we offer insights that have implications for both media profitability and the impact of advertising on consumers.

Keywords: *advertising, media, information design, dynamic attention, attention steering, two-sided market*

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1. Introduction

Advertising is often disseminated through media. Historically, the evolution of mass media has advanced the formats of advertising. Printed advertisements emerged with the invention of printing technology over a thousand years ago.[1] This advertising format thrived for centuries until the advent of electronic broadcasting technologies in the early 20th century. Since then radio and television commercials gradually became the primary advertising tools for marketers and significant revenue sources for media.[2] In the late 20th century, the media landscape underwent a disruptive transformation with the emergence of the internet, marking the third revolution of media advertising. Digital advertising quickly became indispensable for marketers.[3]

Drawing upon the evolution of advertising media, this study seeks to elucidate the role of media format in advertising, through an examination of the fundamental distinctions among the evolving media formats. Such an inquiry can offer pertinent managerial implications. Notably, the rise of digital media does not render traditional media obsolete; rather, many media platforms today incorporate a variety of advertising formats, many of which share fundamental similarities with traditional ones. Hence, which media format to adopt and how to manage it effectively become important questions. YouTube, for example, features a range of ad formats including display ads, skippable and non-skippable video ads. Amazon similarly provides diverse advertising formats — video, audio, and display ads — across its various platforms. These media platforms must decide, or delegate the decision to creators or advertisers decide, on the appropriate format for different contexts.

To make progress, it is imperative to integrate the various media formats into a unified modeling framework. Accordingly, we develop a continuous-time model grounded in the premise that advertisements seek to capture consumer attention, a process that unfolds over time. In particular, we posit that an advertisement can inform a consumer about the existence of a product and its match to her. The match signal, however, only

[1] The first known movable-type printing system, developed by Bi Sheng around 1040 C.E. during the Northern Song dynasty (960-1279 C.E.), paved the way for printed ads in China. The known example is a print poster made by a needle shop in the Ji'nan city. Johannes Gutenberg's introduction of mechanical movable-type printing around 1439 C.E. marked the onset of modern print media.

[2] The first paid radio commercial was aired on AT&T-owned New York station WEAF in 1922, promoting the apartments in Jackson Heights owned by Queensboro Corporation. The first legal television commercial was broadcast on NBC's WNBT-TV in 1941, featuring Bulova's watch for ten seconds.

[3] The first banner ad appeared on Hotwired.com in 1994, sponsored by AT&T.

arrives at a random time when the consumer pays attention. Media formats differ in their capabilities of shaping this attention process. Given a particular format, the consumer can decide at any moment what to attend to: the advertisement, the media content, or the outside option, provided they are available. It is likely that after a period of time, the consumer may have learned nothing about the match, resulting in no conversion. None of the parties possesses superior information about the match value *ex ante*. If the match signal does arrive, this event is only known to the consumer, but not to the medium or the advertiser.

In a nutshell, the model features both the dynamic attention problem of a consumer and the dynamic attention steering of a medium. These two jointly determine the equilibrium behaviors. To keep our analysis focused on these forces, we have abstracted away from a variety of factors that could affect advertising effectiveness — to name a few, advertised price, brand name, product category, ad copy, scheduling, and targeting.

At a high level, the model chronicles the evolution of advertising media, with each wave of media revolution corresponding to different degree of control over consumers' attention. This has motivated us to classify media formats into three broad types. In the first type, a medium presents its content and advertisement simultaneously, without the ability to compel consumer attention to the ad. The consumer, however, can flexibly determine what to pay attention to. This format is referred to as *static advertising*, and is well-suited for print media such as newspapers, magazines, and online websites with static display ads. In the second type, a medium can choose when to expose a consumer to an ad or to content, but not both simultaneously. This flexibility enables the medium to show the ad exclusively at specific times, as seen in broadcast media such as television and radio. We term this format *sequential advertising*. In the third type, a medium has the option to release either the ad, the content, or both, at any given moment. A notable feature is that the medium can, after displaying the ad for a period, provide a consumer with the choice to skip it. Essentially, this form of advertising offers an interactive element and accordingly is termed *interactive advertising*. Many online media, such as video streaming platforms that permit viewers to skip ads, fall into this category.

In essence, this taxonomy captures different modes of communication between media and consumers, and underlines the important role of the media's ability to steer consumer attention. A key theoretical insight is that the steering capability interacts with the two fundamental problems in media advertising.[4] First, when making the attention decision, the consumer does not internalize the advertising benefit to the medium. Thus, the incentives of the consumer and the medium may not necessarily be aligned. Second, the medium does not directly observe the consumer's private information — whether she has obtained the ad signal when paying the attention.

Under static advertising, a medium presents both its content and an ad simultaneously to a consumer, who then decides how to allocate her attention. Paying attention to the ad activates the signal arrival process. The consumer must decide whether to pay attention to it and for how long. This is an infinite-horizon optimal stopping problem and the optimal solution to it follows a simple threshold strategy. The consumer pays attention to the ad whenever the content value falls below the threshold λv_c, where λ captures how soon the match signal arrives (i.e., the informativeness of the ad) and v_c is the consumer's expected surplus conditional on a match. She keeps her attention to the ad until the signal arrives. At the outset, introducing advertising is optimal to the medium only if the consumer has the interest to pay attention to it. In this case, the incentives of the consumer and the medium are aligned. However, for a relatively less informative ad, their incentives are not aligned. In the lack of control capability, the medium may experience a loss in advertising revenue.

In the case of sequential advertising, a medium can present an ad and the media content sequentially. A critical question for the medium is how long the exposure should be, an inquiry that can be addressed through the continuous-time framework. Although the medium possesses greater control over the consumer's attention, the lack of private information (i.e., the arrival of ad signal) has led the medium to impose a deterministic time window for the ad exposure. The consumer then faces a *finite-horizon* control problem, typically intractable in continuous time. However, due to the exponential arrival assumption, the solution becomes tractable, mirroring that of its infinite-horizon counterpart. Anticipating the consumer's attention behavior, the medium's optimal policy is shown to be a simple function of the expected match values of advertising, the content value to the consumer, and the informativeness of the ad. If the ad is moderately informative, unlike the case in static advertising, the medium can still profit from advertising despite the misalignment of incentives.

[4] We thank the anonymous reviewer for pointing this out.

Unlike static advertising, where the medium has "too little" control, and sequential advertising, where the medium has "too much" control, interactive advertising strikes a balance between the two. In this case, a medium can simultaneously present an ad and its media content at any time, granting the consumer the flexibility to choose between them. As a result, the medium faces a decision regarding whether and when to introduce this option. If the ad is not too informative, then the consumer has no interest in paying attention to the ad even if it is available. This compels the medium to introduce a *minimum run time* to keep the consumer engaged with the ad. Then the medium is faced with the same problem under sequential advertising — when to terminate the ad's run time. In contrast, if the ad is sufficiently informative, then the consumer is more willing to pay attention to it. It is then optimal for the medium to introduce the skip option as early as possible, akin to the situation in static advertising.

Building on these characterizations, we explore the relevant managerial and welfare implications in Section 4. Regarding the managerial implications, as expected, an increase in the informativeness of an ad can consistently enhance the profitability of all media types. This implies that improving targeting accuracy not only benefits digital media but also traditional media in terms of media profits. Furthermore, our analysis underscores how the interactive feature has provided digital advertising an additional edge over traditional advertising. This holds true only when ads are sufficiently informative, because then interactive advertising can generate strictly higher profits than either the sequential or static advertising formats. Nonetheless, it is conceivable that both sequential and static advertising can perform equally well as interactive advertising under certain conditions. Specifically, when the ad is highly informative, interactive advertising is essentially equivalent to the static format. This aligns with common practices, where some skippable ads allow consumers to skip them from the outset. However, when an ad is moderately informative, interactive advertising becomes equivalent to sequential advertising. This explains why certain skippable ads entail a minimum run time. In essence, these equivalence results shed light on when skippable ads should be *completely* skippable or *partially* skippable.

More broadly, these insights can offer guidance to multi-format media platforms that need to decide which advertising format to adopt based on their technological capabilities. At the same time, as technologies continue to evolve, new advertising formats continually emerge. The model can help us understand how a new format is connected to existing ones. This understanding is also relevant to entrepreneurs or investors who aim to assess the profitability of new advertising technologies.

In terms of welfare implications, we find that, as an ad becomes more informative, it does not always benefit consumers. Under sequential advertising, consumers can become worse off if the ad becomes more informative. This is because the more informative ad incentivizes the medium to extend the ad exposure time. The same force applies to interactive advertising as well when it is equivalent to sequential advertising (i.e. in the format of partially skippable ad). One implication of this finding is that, if improved targeting technologies enhance the informativeness of an ad, then the ad can result in greater consumer reactance. Indeed, Goldfarb and Tucker [2011] find that when intrusive online ads are more targeted, they become less effective. However, our analysis further suggests that this result is not absolute: as an interactive ad becomes sufficiently more informative, consumers can skip the ad entirely and thus be less likely to trigger consumer reactance. These results collectively explain the consumer heterogeneity in ad annoyance across different media formats, and why digital ads, particularly those with the intrusive nature, are less welcome by consumers, even when they are targeted.

We next review the relevant literature. Section 2 sets up the model for our analysis. Section 3 characterizes the equilibrium behaviors under different media formats. Building on these characterizations, Section 4 discusses the implications for media profits and consumer welfare. In Section 5, we explore a few directions that the model can be extended. Section 6 concludes.

Related Literature

There is a vast literature on advertising, and much has been focused on the persuasive, informative, and complementary effects of advertising.[5] However, relatively few efforts have been devoted to understanding how advertising communication may be endogenously influenced by the type of media in which the ads are embedded. More recently, researchers have begun to study media advertising through the lens of two-sided markets (e.g., Armstrong 2006, Rochet and Tirole 2006, Weyl 2010) – that is, media platforms seek to attract both advertisers and consumers and benefit from their interactions. Much of the focus has been on the pricing problem, particularly on how pricing

[5] A comprehensive review here is unnecessary and beyond the scope of this paper. There are several excellent reviews with different focuses (e.g., Bagwell 2007, Renault 2015).

policies are shaped by the cross-side externalities, and whether market provision of advertising is efficient. In contrast, the focus of this paper is on the information design problem of media platforms, a problem that has received relatively little attention. At the same time, our model can micro-found the externality of advertising to consumers that has been assumed in the extant work on two-sided markets (e.g., Dukes and Gal-Or 2003, Dukes 2004, Anderson and Coate 2005, Anderson and Gans 2011).

The externality of advertising to consumers is, to a large extent, motivated by the phenomenon of ad avoidance. This has been particularly relevant for sequential advertising. Television viewers are known to adopt various ways to avoid commercials. For example, they can switch channels (Siddarth and Chattopadhyay 1998) or perform other tasks (Wilbur 2008). Zhou [2004] presents an early theoretical analysis on how viewers' avoidance behavior affects the choice of commercial breaks. In order to study both the frequency and length of commercial breaks, his model makes a convenient assumption that some viewers drop out of a TV program when a commercial break begins and once they leave the program, they never come back. This has led to the prediction that the TV program will never start with a commercial. We instead explicitly model a viewer's optimal decisions over time given the possible options at hand. Our analysis illustrates the possibility of a TV program to show an ad upfront before showing the program content. More recently, the incentive of consumers to switch away from ads has fueled the growth of ad-avoidance technologies, such as the TiVo digital video recorder and many ad-blocking softwares or mobile applications (Todri 2022). These technologies can remove ads entirely from media platforms, posing a threat to the ecosystem of the media market (Anderson and Gans 2011, Johnson 2013, Shiller et al. 2018, Gritckevich et al. 2022). One solution to the problem is to leverage the power of price discrimination by offering consumers a choice between ad-funded content and ad-free content at a higher price (Lin 2020). Our analysis of interactive advertising also suggests that leaving more control to consumers could be beneficial to a media platform.

Our theory highlights *interactivity* as one of the key features of digital advertising. Although this feature is naturally brought by the Internet technology, it has not gained much attention until recently when the so-called "skippable ads" become increasingly popular. YouTube's *TrueView* ad is a leading example. Dukes et al. [2022] present an early theoretical analysis of why and when skippable ads can be profitable to a media platform. They argue that the skippable format can encourage a platform to serve more ads, thereby exposing consumers to more beneficial transactions while not irritating them. This indirect demand-enhancing effect can eventually benefit the platform. We focus on different research questions and thus tailor a different model to answer them. In particular, we are interested in the fundamental differences among various types of media formats and how they affect consumer decision and welfare. We therefore allow for a richer media strategy space such that under interactive advertising, a media platform can determine at any point when to introduce a skip option. As a result, interactive ads (i.e., digital ads) can be either partially skippable, entailing compulsory view time just as sequential ads, or completely skippable, without forcing consumers to view an ad just as static ads.[6] Our continuous-time model provides a tractable framework for such an analysis. It further allows us to examine how the choice of media format can depend on the informativeness of an ad, which is not the focus of Dukes et al. [2022].

The optimal advertising design problem we study is also related to a traditional literature on the dynamic advertising allocation problem, which has a long history in marketing, economics, and operations research (e.g., Vidale and Wolfe 1957, Nerlove and Arrow 1962, Little 1979, Mahajan and Muller 1986). Sethi [1977] and Feichtinger et al. [1994] provide a comprehensive review of this line of research. Although our theory similarly builds on a continuous-time control problem, the approach taken is rather different. We explicitly model the interaction process between advertisers and consumers, instead of assuming an aggregate model that relates advertising spending to product sales. Thus, while the literature speaks to the problem of scheduling ads over long periods (e.g., weeks or months), our model focuses on the allocation of an ad in a short period. Furthermore, existing studies are largely motivated by the problem of advertising spending on television, and thus are unable to provide guidelines for advertising in the digital age, which features both targetability and interactivity. The theory developed in this study fills this gap and provides insight into how various media formats influence advertising. Methodologically, the proposed theory solves both the media and consumers' control problems simultaneously, in contrast to studies that focuses on the firm's problem without a micro model of consumers.

The dynamic attention design problem studied here also connects two streams of research in information economics. The first stream investigates how individuals (e.g.,

[6] In Dukes et al. [2022], skippable ads are always partially skippable, and thus they focus on comparing partially skippable ads to traditional ads in the sequential format.

consumers, job seekers) actively acquire information over time, but assumes information is exogenously given. The seminal papers by McCall [1970] and Weitzman [1979] have inspired extensive work on optimal information search. The second stream focuses on how an individual (sender) persuades another (receiver) to change her actions within a symmetric information framework but assumes that receivers passively update beliefs given the information provided by senders. The static Bayesian persuasion framework was first introduced by Kamenica and Gentzkow [2011] and has subsequently been extended to dynamic settings in which senders can control the flow of information over time (e.g., Ely et al. 2015, Ely 2017). However, even in these dynamic settings, receivers are often assumed to be myopic with no dynamic incentive to acquire information. This study integrates both research streams in the analysis of a media market: media control the flow of information with the intention of influencing consumers, who in turn have dynamic incentive to acquire information.

2. The Model

We model a simple market with one medium, one consumer, and one advertiser. The goal is to provide a simple framework that allows us to better understand the role of media in advertising and how it varies across different formats. Depending on its technological capability, the medium can steer the consumer's attention over time. Recognizing this, the consumer decides what to pay attention to and how. We next describe each of these players' decision environment in detail.

2.1 Medium

The medium can produce media content M continuously over an infinite time horizon $T = \infty$, bringing informational or entertainment value to the consumer. In reality, media content can take a variety of forms, such as news articles, television programs, online videos, music streaming, news feeds on social media, etc. In general, the flow utility of the consumer's content consumption may vary over time. Without loss of generality, we assume that the consumer values the media content at a constant rate: $m(t) = m \geq 0$ for any time t.[7] The production cost of the media content is assumed to have been sunk

[7]Section 5 discusses an extension of the model with time-varying content.

and normalized to be zero. Our focus is on the allocation decisions after the media content becomes available.

In addition to media content, the medium can expose the consumer to an ad A sponsored by the advertiser. In theory, at any time t, the full set of options for the medium's decision $d(t)$ is $\mathcal{D} = \{A, M, \{A, M\}\}$, where the subset $\{A, M\}$ implies that the medium allocates both advertising and media content simultaneously to the consumer.[8] In practice, however, there are technological constraints on the medium's actions. Thus, the medium's actions are often confined to a subset of $\mathcal{D}$. This observation suggests an approach that classifies advertising based on the media formats.

Definition 1 (A Taxonomy of Media Advertising Formats):

1. ***Static Advertising***: $\mathcal{D}_{Static} = \{\{A, M\}\}$. *The medium can only run the ad and the media content simultaneously. The consumer can choose to process one of the two. Example: print media ads.*
2. ***Sequential Advertising***: $\mathcal{D}_{Sequential} = \{A, M\}$. *The medium can run either the ad or the content in any sequence but cannot run both simultaneously. Example: television and radio commercials, online video ads.*
3. ***Interactive Advertising***: $\mathcal{D}_{Interactive} = \{A, M, \{A, M\}\}$. *The medium can run the ad and the content either simultaneously or sequentially at any point in time. Example: online skippable ads like YouTube's "Trueview" ads, swipeable carousels ads.*

As will be shown in the following analysis, this taxonomy helps us to understand the fundamental differences and connections between different advertising formats. It is worth pointing out that ads on many online or digital media platforms are not necessarily interactive. For example, many websites display simple static banner ads that are combinations of text and images. YouTube, in addition to allowing advertisers to adopt the skippable TrueView ads, offers alternatives such as non-skippable in-stream ads or bumper ads. The variety of ad formats on digital media calls for a deeper understanding of their nature, which is precisely the focus of the main analysis.

[8] Here, when both the ad and media content are bundled, it does not mean that the consumer will process both. Instead, she needs to choose which one to process due to limited attention, a point discussed in more detail below.

2.2 Consumer

Whenever the consumer decides to pay attention to the running ad, the ad information process is activated. This information process must be distinguished from the ad exposure process, as exposure does not guarantee that the consumer will pay attention to the ad. Thus, it is useful to define the ad process time x, which is the time the consumer takes to process the ad. Note that this process time is endogenously related to the universal timeline t and hence the function $x(t)$ will be used in the analysis. The role of advertising is both to inform the consumer of the existence of the advertised product (or the advertised brand more generally) and to provide her with information about whether the product is a match to her.

This match information, or the signal, is stochastic in two dimensions. First, it arrives at a random time $\sigma \geq 0$, following an arrival distribution $F_\sigma(x) \equiv Pr(\sigma \leq x)$. To keep the analysis tractable, the signal arrival process is assumed to follow an exponential distribution with parameter λ.[9] This parameter captures the *informativeness* of the ad: for a fixed time period, an ad with a higher value of λ is more likely to produce the signal. Second, upon arrival, there is a probability $\rho > 0$ the signal tells the consumer that there is a match, and a probability $1 - \rho$ that the signal indicates that there is no match. Conditional on a match, the one-off benefit to the consumer is $\tilde{v}_c \geq 0$ and to the advertiser is $\tilde{v}_a \geq 0$.[10] We assume that if the signal does not arrive and thus the consumer remains uninformed about the value of the advertised product, the consumer will not make a purchase.[11] To simplify notation, let $v_c \equiv \rho\tilde{v}_c$ and $v_a \equiv \rho\tilde{v}_a$ denote the expected ad values to the consumer and the advertiser conditional on the signal arrival. Unlike the signal arrival process that speaks to the efficiency of the ad, the realization of the match value captures the effectiveness of the ad. Further, the arrival of the signal and the realization of the match value are independent.[12] The setup of the information

[9] In modeling arrival processes, the exponential distribution (or more generally Poisson distribution) is a standard assumption (see for example, Ely 2017). As we will show later, the memoryless feature of this assumption can simplify the analysis substantially.

[10] These benefits can be interpreted as the surplus generated by the transaction between the consumer and advertiser. However, it is likely that consumers may continue to explore an advertised product before a purchase (Mayzlin and Shin 2011, Dukes et al. 2022). Then $\tilde{v}_c$ and $\tilde{v}_a$ represent the total expected value of continuing searching.

[11] This can occur if the prior belief about the product value is sufficiently low, or if the alternative way of acquiring information is too costly.

[12] This assumption avoids the complication that consumers may infer product match based on the mere fact that a targeted ad is delivered to them. See a recent analysis by Shin and Yu [2021] building on that logic.

environment allows us to keep the decision problems of both the consumer and the medium tractable.[13]

It is assumed that at any point in time the consumer can access to an outside option B, which captures any option available other than the media content or the ad. This option may generate reward (possibly stochastically) to the consumer over time. Examples of an outside option include a bathroom break while watching a television program, chatting with a friend while reading a newspaper, or checking emails while watching an online video. This outside option can be viewed as the opportunity cost of paying attention to the ad. As such, it can be interpreted as a search cost.[14] To keep the analysis simple, it is assumed that this option yields a constant reward $b(t) = b$. To avoid triviality, we assume that $b < m$. In later part of the analysis, b will be further normalized to zero.

At any point in time, the consumer chooses an attention action $c(t)$ from a choice set $\mathcal{C}(t)$ that is partially determined by the medium. Formally, $\mathcal{C}(t) = \{B, d(t)\}$. For example, if the medium runs the ad at time t, $d(t) = A$, then the consumer can choose to either pay attention to the ad (i.e., $c(t) = A$), or take the outside option (i.e., $c(t) = B$). In both cases, she does not consume the media content. If, however, $d(t) = \{A, M\}$, then she has the additional option of skipping the ad and consuming the content (i.e., $c(t) = M$).

Realistically, paying attention to either the content or the ad demands cognitive efforts. For ease of exposition, we assume these costs to be identical for both the media content and the ad, and normalize them to be zero. In an extension, we show how to incorporate the differential cognitive costs to enrich the model. Nevertheless, as noted above, incorporating the outside option can capture the opportunity cost of the consumer's attention, which is the primary focus of the paper.

The consumer discounts future rewards. The standard approach is to assume that the present value of a future reward at time t is discounted exponentially by a factor of e^{-rt}. Here we adopt the interpretation that the discounting is due to a random termination of the problem which, if occurs, results in no ad processing and content consump-

[13] In a more general and realistic setup, one may allow an ad to generate signals multiple times over time. However, such a model can easily become intractable. As shown later, the consumer is faced with a finite-horizon dynamic problem. In general, the optimal solution to such problems is hard to solve in closed-form.

[14] As argued in Nelson [1974], the marginal cost to a consumer looking at an ad is primarily a time cost. "This time cost will vary by the alternative use of the time used in watching the advertisements." (p.745, Nelson 1974)

tion. This stochastic approach simplifies the analysis because all players share the same discount factor, which leads to a simple formula for the optimal ad allocation.

2.3 Timing and Information Structure

We conclude the model by summarizing the timing and information structure of the game. At any time point t, the medium of format k chooses the action $d(t) \in \mathcal{D}_k$. The consumer observes the medium's decision $d(t)$, and then decides her attention action $c(t) \in \mathcal{C}(t)$ based on the information she has accumulated as of time t.

Under both static and sequential advertising, the medium does not observe $c(t)$ and therefore can not condition its decision on the consumer's action. This information structure aligns well with the reality — newspapers, magazines, television and radio producers are typically unaware of whether a particular consumer has paid attention to an ad. Given this information structure, the medium's action at time t is independent of the history of the consumer's actions prior to time t. However, under interactive advertising, the medium can partially observe a consumer's decision if it presents her with the choice between an ad and media content. Consequently, there is room for the medium to condition its policy on the consumer's action.

Regardless, the medium does not observe the arrival of an ad signal. This assumption is reasonable, considering that the way consumers process an ad is largely concealed and unknown to others.[15]

An equilibrium solution requires simultaneously solving both the medium's allocation problem and the consumer's attention problem. The equilibrium behaviors, naturally, hinges on the media format. The subsequent section is devoted to characterizing these equilibrium behaviors across the three distinct advertising formats.

3. Consumer Attention and Media Policy

[15]Even when a consumer has clicked on an ad, it may not reveal whether she has already acquired the match information relevant to her.

3.1 Static Advertising

Under the static format, the consumer can decide freely whether to pay attention to the ad or to the media content.[16] At the same time, the medium has limited capability to divert the consumer' attention to the ad. The medium's policy, once determined at the outset, remains fixed over time. The question then is: when is the consumer in the mood for ad? Put differently, what is the optimal strategy of the consumer given this information environment? Analyzing this problem also lays the foundation for analysis of more complex advertising models. Lemma 1 below presents the result.

Lemma 1 (Consumer Attention under Static Advertising):

1. *If $\lambda v_c \leq m$, the consumer pays attention to the media content only.*
2. *If $\lambda v_c > m$, the consumer pays attention to the ad first and consumes the content after the ad signal has arrived.*

As the consumer has access to both the outside option and the media content at any decision time, and given that $b < m$, she will never choose the outside option. The consumer's problem is then to choose between attending to the ad A and consuming the media content M. Because the content consumption yields constant flow utility, this problem is an optimal control problem, or more generally, a two-armed bandit problem with a safe arm M and an uncertain arm A.[17] We can leverage the well-established solution to this problem, known as Gittins Index (Gittins 1979). Let us define the index of the ad process, before the arrival of the match signal, as a function of the state $x(t)$, which is the process time of the ad (i.e., the cumulative time the consumer has paid attention to the ad up to a specific point):

$$G_A(x) \equiv \sup_{t>x} \frac{v_c \int_x^t f(s)e^{-rs}ds}{\int_x^t [1 - F(s)]e^{-rs}ds} = \lambda v_c. \tag{1}$$

[16] We have assumed an infinite horizon for static advertising, implying no time limit for the consumer's attention problem. However, many media such as newspapers do have limited space for media content, which is better captured by a finite horizon. Nevertheless, as we show in the case of sequential advertising in Section 3.2, assuming infinite horizon does not alter the solution to the consumer's attention problem when the horizon is finite.

[17] In the bandit problem literature, an "arm" refers to an independent (dynamic) control process. The uncertain arm A defined in the model is equivalent to a *job process* in the job scheduling literature in operations research.

The numerator of Equation 1 captures the the expected reward of exercising the option A continuously up to time t, whereas the denominator measures the expected time of this process. This index value can be understood as how much the consumer is just willing to pay attention to the ad (versus taking the fixed reward $G(x)$).[18] The equality of the above equation follows from the distributional assumption that F is exponential. Thus, the index value does not depend on the process time as long as the signal has not arrived, simplifying the solution: the consumer processes the ad if only if the ad index is greater than the reward of the media content, $\lambda v_c > m$.[19] Once the signal has arrived, the uncertainty is resolved and thus the consumer focuses on the content thereafter. The above reasoning establishes Lemma 1.

We now examine the medium's policy, which boils down to whether to allow advertising at all at the outset of the game. If $\lambda v_c \leq m$, following the optimal strategy, the consumer expects a total value of

$$V^*_{Static} = \int_0^\infty m e^{-rt} dt = \frac{m}{r}. \tag{2}$$

As the medium extracts all the consumer surplus, its profit is $\Pi^*_{Static} = V^*_{Static} = m/r$. In contrast, if $\lambda v_c > m$, the consumer's total expected value following the optimal strategy is

$$V^*_{Static} = \int_0^\infty (v_c + \frac{m}{r}) e^{-rt} dF(t) = \frac{\lambda(r v_c + m)}{r(\lambda + r)}. \tag{3}$$

Collecting both consumer and advertiser surplus, the medium obtains

$$\Pi^*_{Static} = \int_0^\infty (v_c + v_a + \frac{m}{r}) e^{-rt} dF(t) = \frac{\lambda(r(v_c + v_a) + m)}{r(\lambda + r)}. \tag{4}$$

To profit from advertising, the medium needs to ensure that the profit expressed in Equation (4) is greater than the ad-free profit m/r. This requires that $\lambda(v_c + v_a) > m$, a condition that can be satisfied as long as $\lambda v_c > m$. The optimal policy for the medium can then be summarized as follows.

[18] Hence, in solving for the index value $G(x)$, the consumer finds the optimal stopping time that makes her just indifferent between A and a fixed reward $G(x)$.

[19] This simplified threshold is due to the memoryless property of the exponential distribution — if a signal has not arrived, regardless of how long the consumer has waited, the likelihood of obtaining the signal remains unchanged. Hence, the optimal stopping problem remains the same as long as the signal has not arrived. The exponential distribution, however, is a sufficient but not necessary condition for this simplified rule to arise. If the index is nonincreasing with probability one, then the simple threshold strategy continues to apply.

Proposition 1 (Media Policy under Static Advertising): *Under Model $\mathcal{D}_{Static}$, the optimal media policy is to offer the content with the ad if $\lambda v_c > m$. Otherwise, it is optimal to turn off advertising.*

Remark 1. For the consumer to engage with the ad and for the medium to profit from it, the ad must be sufficiently informative and valuable, $\lambda v_c > m$.[20] In this case, the incentives of both the consumer and the medium are perfectly aligned. Therefore, even without possessing the information about signal arrival, the medium can allow the consumer to control her attention in a manner that is desirable to the medium as well. However, for a slightly less informative or less valuable ad that satisfies $\lambda v_c < m < \lambda(v_c + v_a)$, the incentives of both the consumer and the medium are not aligned. This is more likely to occur as v_a becomes larger. While the medium desires the consumer to pay attention to the ad, the consumer prefers to skip it. Because the consumer has full control of her attention, the medium is unable to force her to attend to the ad and must forego the ad altogether. It becomes evident that sequential advertising, introduced subsequently, can alleviate this problem.

3.2 Sequential Advertising

Within this model class, media exert more control over consumer attention. Specifically, at any point in time, a medium can opt to either run an ad or offer the media content, but not both simultaneously. Consumers' attention decisions therefore hinge on how the medium allocates these options. Next, we first analyze the optimal strategy of a consumer under any arbitrary policy, and then solve for the optimal media policy.

The sequential nature of the media format raises the issue of a commitment problem. However, because the medium does not observe the consumer's action, the commitment problem becomes irrelevant here. Hence, the media strategy has the simple form of history-independent allocation of advertising. Consider an arbitrary allocation policy π where a medium runs an ad for a total duration of $S = \int \mathbb{1}\{d(t) = A\}dt$.[21] Note that the class of policies with the same ad duration S is quite large. A policy might stip-

[20] Note that the magnitude of the ad value v_c needs to be weighted by the informativeness λ when compared with the magnitude of content quality m.

[21] Note that the consumer does not observe S at the beginning of the game, but she can rationally expect S in equilibrium. As will be shown, the optimal attention strategy does not rely on whether S is observed or not.

ulate that the ad is interrupted at some point in time and then resumes later. Lemma 2 summarizes the optimal strategy of the consumer.

Lemma 2 (Consumer Attention under Sequential Advertising):

1. *If $\lambda v_c \leq b$, the consumer always chooses the outside option when the ad is running, and consumes the content whenever it is available;*
2. *If $\lambda v_c > b$, the consumer pays attention to the ad when it is running until the match signal arrives or the ad terminates, whichever comes first; she consumes the content whenever it is available.*

Given any arbitrary allocation policy π with ad exposure time S (possibly infinite), let us consider the reduced problem π_0: the ad is run *continuously* from the start for a duration of S without interruption, followed by the provision of media content thereafter. That is,

$$\pi_0 : \quad d(t) = \begin{cases} A & \text{if } t < S; \\ M & \text{if } t \geq S. \end{cases}$$

Once again, the consumer faces a bandit problem akin to that in static advertising, but with two notable distinctions. First, there exists a fixed time window S for the ad process. Second, the media content is unavailable when $t < S$ but becomes the only available option when $t \geq S$. These two features together suggest that the problem can be decomposed into two parts: the first part involves an optimal control problem with $c(t) \in \mathcal{C}(t) = \{A, B\}$ for $t < S$, and the second part simply entails choosing the media content, that is, $c(t) = M$ for all $t \geq S$.

The first part is a *finite-horizon* bandit problem, under which the Gittins Index strategy is not guaranteed to be optimal. However, under the assumption that the ad process follows an exponential process, the flow payoff of paying attention to advertising is non-increasing with probability one, which satisfies the *deteriorating arm* condition (Weber 1992). Then the index strategy remains optimal under a finite horizon.[22] Following the

[22]Intuitively, in solving the index value as defined in Equation 1, the consumer finds it optimal to stop immediately after the next infinitesimal time interval. That is, the optimal stopping time $t \to x$. This implies that the time horizon appears to be irrelevant in her decision. She acts as if she is myopic despite her forward-looking incentive. Note that the exponential distribution is a sufficient but not necessary assumption for this result to hold. As long as the arrival distribution can lead to the non-increasing index function, then we can continue to use the simple threshold for the finite-horizon problem.

index strategy, the consumer chooses to process the ad if and only if $\lambda v_c > b$.[23]

Note that the original problem under policy π can be viewed as introducing interruptions to the ad process given in the reduced problem π_0, while keeping the total ad exposure time S fixed. During these interruptions, i.e., when the ad is not running, the consumer's only option is to consume the media content. These interruptions neither affect the state (process time) nor the payoff of the ad process. Thus, following the index strategy for problem π_0 is optimal for the original problem π, establishing Lemma 2.

Given the optimal strategy of the consumer, the medium can choose an optimal allocation policy to maximize profit. In this simple model, the medium can extract all the value generated by a match between the consumer and the advertiser by charging the consumer a subscription fee and charging the advertiser an advertising fee. However, the medium cannot base its fee on whether the consumer actually pays attention to the ad, because the attention is unobservable and thus it is not a feasible basis for a contract (Rochet and Tirole 2006). Essentially, the medium is maximizing the total welfare of all parties. The following proposition summarizes the result.

Proposition 2 (Media Policy under Sequential Advertising): *Under Model $\mathcal{D}_{Sequential}$,*

1. *if $\lambda v_c \leq b$, the medium does not run the ad; and*
2. *if $\lambda v_c > b$, the optimal media policy is to run the ad continuously from the start for a period of $S^* = \max\{\frac{1}{\lambda}\ln\frac{\lambda(v_c+v_a)}{m}, 0\}$ and to present the media content thereafter.*

The first part follows straightforwardly from Lemma 2. To prove the second part, note that if the medium chooses A at time t, then Lemma 2 implies that its flow payoff is $(v_c + v_a)f(x(t))$. If the medium chooses M at any time, the flow payoff is always m. Note further that if the medium suspends the ad at some time t and then resumes the ad at a later time $t' > t$, the state does not change. Then the medium's problem is an optimal stopping problem: if it is optimal to choose M at time t, it will continue to do so thereafter. Hence, the optimal policy takes the form of continuously running the ad

[23] Note that the decision problem during the ad exposure period $t < S$ is slightly different from that under static advertising. In the latter case, the ad is already available for the consumer from the start. In the sequential advertising case, however, the ad content is sequentially released by the medium, regardless of whether the consumer is paying attention. Recall that we distinguish the ad information process from the ad exposure process. The fact that the ad is displayed does not guarantee that the consumer has paid attention to it. The ad information process is activated only if the consumer pays attention. Hence, when the consumer takes the outside option and does not pay attention to the ad, the ad process is paused. The state (ad process time, not exposure time) does not change during the break.

for a period of S and then presenting the content thereafter. The optimal S must solve:

$$\max_{S\geq 0} \Pi_{Sequential}(S) = \int_0^S (v_c + v_a)e^{-rs}dF(s) + \int_S^\infty me^{-rs}ds. \tag{5}$$

The first-order condition is given by

$$\lambda(v_c + v_a)e^{-(\lambda+r)S} - me^{-rS} = 0. \tag{6}$$

It immediately follows that the interior solution is[24]

$$S^* = \frac{1}{\lambda}\ln\frac{\lambda(v_c + v_a)}{m}, \tag{7}$$

which is positive if $\lambda(v_c + v_a) > m$. Otherwise, the corner solution $S^* = 0$ applies. We thus have proved Proposition 2.

Intuitively, the optimal ad exposure time S^* strikes a balance between the values generated by the consumer-advertiser match and the opportunity cost of delaying content provision to the consumer. The outcome that the medium prefers to show the ad before introducing the media content aligns with common practices in TV commercials and pre-roll ads featured on many online platforms (e.g. YouTube). The optimal ad exposure time is a simple function of the market parameters, allowing us to make sharp predictions about how these parameters influence the time windows for advertising. The following proposition characterizes these relationships.

Proposition 3 (Comparative Statics under Sequential Advertising): *The optimal exposure time of sequential advertising*

1. *weakly decreases with the content quality* m*,*
2. *weakly increases with the expected match values for the consumer and advertiser* (v_c, v_a)*, and*
3. *weakly increases as the ad informativeness* λ *increases if* $\lambda < em/(v_c+v_a)$*, but weakly decreases in* λ *if* $\lambda > em/(v_c + v_a)$*.*

The first two parts of Proposition 3 straightforwardly follow from the solution in Equation 7 and are fairly intuitive. If the media content becomes more attractive to

[24]It can be easily verified that the second-order condition is also satisfied.

the consumer, then the medium should reduce the exposure time of the ad. However, if the total value brought about by successfully matching the consumer and advertiser is higher, then the consumer should be exposed to the ad for a longer period to increase her chance of obtaining the match signal.

The third part of the proposition is obtained by examining the derivative of the interior solution S^* with respect to λ,

$$\frac{\partial S^*}{\partial \lambda} = \frac{1}{\lambda^2}\left[1 - \ln \frac{\lambda(v_c + v_a)}{m}\right], \tag{8}$$

which is positive if $\lambda < em/(v_c + v_a)$, and negative if $\lambda > em/(v_c + v_a)$. This result illustrates the nonmonotonic relationship between the optimal exposure time for advertising and the informativeness of the ad captured by λ. Figure 1 provides an illustration. Because λ is the hazard rate of the exponential distribution, the probability of observing the signal, given no signal before time x, is $\lambda = f(x)/(1 - F(x))$, $\forall x$. If λ is very small, an increase in λ can effectively increase the chance of observing the signal. This motivates the medium to run the ad for a longer period. Conversely, if λ is sufficiently large, the ad signal arrives much sooner. In that case, it is more important for the medium to reduce the ad exposure time so that it can introduce the content sooner. It is worth noticing that both Propositions 2 and 3 are conveniently obtained because of the continuous-time setup.

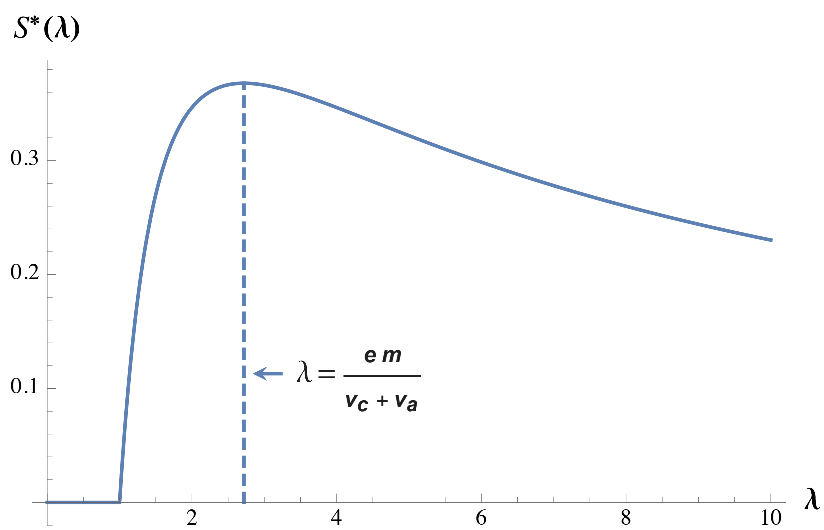


Figure 1: Optimal Ad Exposure as a Function of Ad Efficiency ($v_c + v_a = 1$, $m = 1$)

Remark 2. Recall that static advertising suffers from the incentive misalignment problem when $\lambda v_c < m < \lambda(v_c + v_a)$ (see Remark 1). This problem is alleviated under sequential advertising. In particular, the medium now can steer the consumer's attention toward the ad, even though she prefers to enjoy the media content. Yet, the ability to steer the consumer's attention has an undesirable consequence: because of the lack of private information (i.e., the arrival of the match signal), the medium has to incur inefficiency when determining the ad exposure window. This problem becomes pronounced for a very informative ad $\lambda v_c > m$, under which it would be optimal for the consumer to terminate the attention to the ad only when the signal has arrived. Next, we demonstrate how interactive advertising can resolve this issue.

3.3 Interactive Advertising

Under the interactive advertising model, the medium has one additional control option $d(t) = \{A, M\}$, which allows the consumer to choose between attending to the ad and consuming the media content. Essentially, interactive advertising integrates both static and sequential advertising by enabling one of them at any time t. Whenever the medium is focusing on sequential advertising, either the ad or the content is running exclusively. Again, when running the ad exclusively, the medium does not directly observe the consumer's action, rendering the commitment issue irrelevant. However, when the medium allows the consumer to choose between the ad and the content, it can observe the decision outcome. For example, an online video streaming platform can allow users to skip an ad to watch a video. By clicking on the skip option, a user reveals to the platform that she prefers to watch the video content instead of attending to the ad. Observing the consumer's action opens the possibility for the platform to condition its policy on her action whenever $d(t) = \{A, M\}$ is introduced.

This commitment problem adds complexity to the analysis. We first establish the connection to the earlier results on the consumer's problem by assuming that the medium can commit to a policy that is independent of the consumer's decision. That is, whenever $d(t) = \{A, M\}$ is introduced, it specifies for how long this option will last before expiration and commits to it.

Lemma 3 (Consumer Attention under Interactive Advertising with Commitment):

1. *When the medium exclusively runs A or M, the consumer follows the strategy under sequential advertising in Lemma 2.*

2. *When the medium introduces the option $\{A, M\}$, the consumer follows the strategy under static advertising in Lemma 1.*

The first part immediately follows by recognizing the equivalence of the consumer problem to that under sequential advertising, as analyzed in subsection 3.2. The second part can be obtained by extending the result in subsection 3.1 to the finite-horizon case and following the same logic adopted in the proof of Lemma 2.

Note that the above result does not fully pin down the consumer's solution because it is restricted to the committed case. However, as we show next, this is indeed the problem that the consumer faces in an equilibrium. The following lemma suggests that the optimal media policy has a simple form that entails commitment.

Lemma 4 (Format of Media Policy under Interactive Advertising): *Under Model $\mathcal{D}_{Interactive}$, the optimal media policy takes the following form:*

1. *the medium runs the ad A exclusively from the start for a period of S_m, followed by granting the consumer the optional control $\{A, M\}$ thereafter, and*
2. *the medium never overturns the consumer's decision.*

PROOF: See the appendix. ■

Lemma 4 suggests that the optimal strategy is characterized by a *minimum run time* S_m that forces the consumer to attend to the ad and a *skip* control that allows the consumer to skip through the ad to consume the media content immediately. This minimum run time S_m is similar to the ad duration S under the sequential format, as both represent time periods of exclusive ad exposure. The "skippable" format is widely adopted in practice. For example, YouTube's skippable video ads entail a mandatory period of five seconds during which an ad must be watched, followed by an option to skip. Another example is swipeable carousel ads which are popular on social media such as Facebook. Users can freely swipe through multiple product images and/or videos provided, or skip all together if they wish. At first glance, such a practice appears puzzling: after all, the medium has already decided to let consumers decide whether to watch an ad depending on their interests. Yet, our analysis reveals why this could arise.

To understand the intuition behind Lemma 4 (i.e., the irrelevance of commitment problem), we need to distinguish two cases. First, if $\lambda v_c \leq m$, then the consumer prefers to consume the media content over the ad when both are present. This holds true regardless of whether or not the consumer has already obtained the ad signal prior to the introduction of the skip option. Hence, even after observing that the consumer has chosen the media content, the medium has no information about the signal arrival. The problem to the medium reduces to an optimal stopping one: when to introduce the skip control, or equivalently, the choice of S_m.[25]

Second, if $\lambda v_c > m$, then the consumer prefers to process the ad over the media content when both are available, as long as the ad signal has not arrived. In this case, the consumer's action can reveal the signal arrival. At the point of S_m when the medium observes that the consumer does not choose A, it can infer that she has already obtained the ad signal, and thus, it is optimal for the medium to let her switch attention to the media content. It has no incentive to force the consumer to attend to the ad. In contrast, when the medium observes that the consumer continues to choose A after the skip option is introduced, it learns that the signal has not arrived. It is then in the interest of both the consumer and the medium to continue processing the ad until the signal arrival. Any strategy interrupting this process is suboptimal.

It is also worth noticing that after the minimum run time S_m, the optimal format is to give the consumer an option to choose between the media content and the ad, rather than simply showing the media content if the consumer skips the ad. In theory, it is possible that consumers may find an ad interesting and prefer to continue to watch it even after the minimum run time. The optimal format allows for this possibility. In practice, media platforms like YouTube can relegate an ad to the side bar as a banner ad, allowing interested viewers can continue watching it.

Based on the simplifying results in Lemma 4, we can readily obtain the optimal media policy under interactive advertising, as summarized in the following proposition.

Proposition 4 (Media Policy under Interactive Advertising): *Under Model $\mathcal{D}_{Interactive}$,*

1. *if $\lambda v_c \leq b$, the medium does not run the ad, and*

[25]The reasoning here may breakdown if the medium learns that the longer the consumer pays attention to the ad, the more likely she is interested it. Then the medium will have an incentive to revise the original ad duration. This requires the assumptions that: (a) the medium observes the consumer's attention behavior so that it can infer the updated belief; and (b) there are multiple signals so that after one signal has arrived, the consumer will still have an incentive to continue watching the ad. Assumption (b) is a very realistic one. However, assumption (a) is very strong because the consumer can always distract away by taking the outside option without being observed by the medium.

2. *if $\lambda v_c > b$, the optimal media policy is to continuously run the ad from the start for a minimum period of $S^*_m = \frac{1}{\lambda} \ln \frac{\lambda(v_c+v_a)}{m}$ if $\lambda v_c < m < \lambda(v_c+v_a)$, and $S^*_m = 0$ otherwise. After S^*_m, the skip option is introduced.*

By Lemma 3, the consumer will pay attention to the ad until the signal arrival or the termination of the ad, whichever comes first. For time $t > S_m$, the consumer chooses to skip the ad and consume the media content if $\lambda v_c \leq m$. Hence, the optimal allocation policy is the same as that under sequential advertising: the medium runs the ad exclusively for a fixed time window S^*_m as long as $m < \lambda(v_c + v_a)$, and $S^*_m = S^*$ in Equation 7.

If $\lambda v_c > m$, the consumer will continue to pay attention to the ad for any time $t > S_m$ as long as the signal has not arrived. In this case, any medium's choice of $S_m > 0$ will be strictly dominated by $S_m = 0$. To see why, suppose $S_m > 0$. If the ad signal arrives at some time $t < S_m$, then the consumer has no access to the media content and thus can only choose the outside option within the time period $[\sigma, S_m]$. This implies that the medium could have improved the profit by letting the consumer choose M within $[\sigma, S_m]$, a contradiction. Hence, it must be that $S^*_m = 0$ in this case.

Comparing the optimal ad duration S^* in Proposition 2 and S^*_m in Proposition 4, the two are the same except when the ad is sufficiently informative or valuable, $\lambda v_c > m$. In this case, the interactive format allows the medium to present both the media content and the ad simultaneously to the consumer. Showing the consumer the ad exclusively does not improve media profit because the consumer finds it optimal to engage with the ad anyway.

Remark 3. Recall that static advertising features "too little control" of the medium, giving the consumer full autonomy to decide what to pay attention to. This can better align the incentives of the consumer and the medium only if the ad is very informative, but otherwise it suffers from incentive misalignment (see Remark 1). In stark contrast, sequential advertising implies that the medium has "too much control" as it offers only one option at a time to the consumer. While sequential advertising can alleviate the loss due to incentive misalignment if the ad is moderately informative, it inevitably leads to efficiency lost for a more informative ad because of the lack of information about the signal arrival (see Remark 2). Interactive advertising resolves the issues by striking a balance between the two extremes of control situation: if the ad is highly informative, then the medium can lift the control, handing it over to the consumer; if the ad is only moderately informative, then the medium can tighten up the control just like in sequential advertising to profit from advertising. Figure 2 illustrates the high-level connections among the three advertising formats.

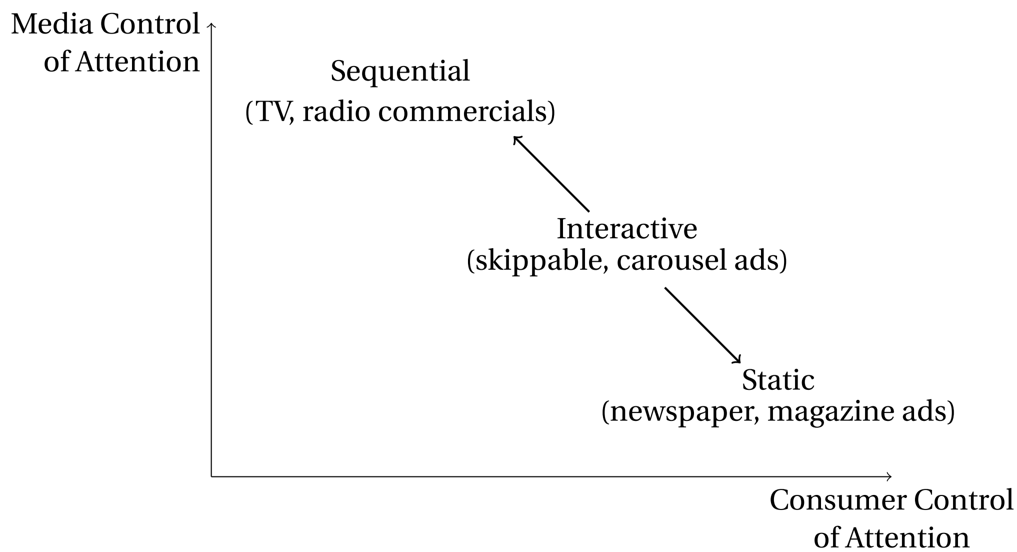


Figure 2: Conceptual Relationships among the Three Ad Formats

4. Implications

The equilibrium characterizations thus far have shed light on the role of media format in advertising and the fundamental differences among various media formats. In this section, we explore the implications for media profitability and consumer welfare, with a particular focus on how the informativeness of an ad can influence the outcomes. Without loss of generality and to streamline the exposition, in this section, the outside option is normalized to zero, i.e., $b = 0$.

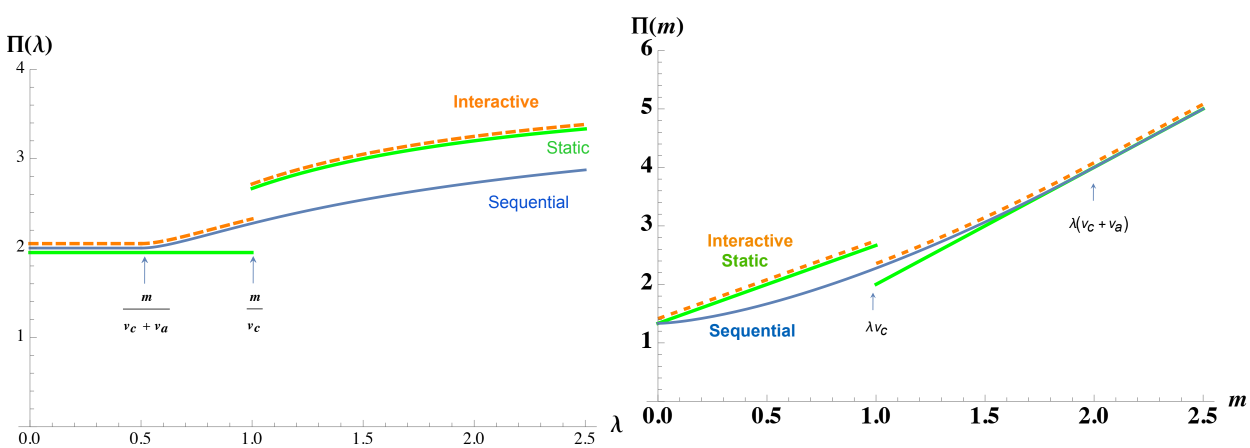
4.1 Media Profitability and Equivalence

Understanding the profitability of a media format has at least two benefits. First, many digital media platforms nowadays can run different formats. It is important for them, or their advertisers or creators, to decide which format to adopt for a particular advertisement. Second, advertisers or entrepreneurs interested in new advertising technologies need to understand how effective it is for a particular media format.

Because interactive advertising admits the richest set of actions for a medium, it is no surprise that this format dominates, at least weakly, the other media formats. However, this conclusion is not immediately obvious. In many games with strategic responses, being more flexible in actions does not lead to a higher profit.[26] Our analysis in Section 3.3 suggests that the medium can act as if it can commit to a particular media policy (i.e., Lemma 4). Hence, the medium can become more profitable under the interactive format. The interesting question then is: under what circumstances can static advertising and sequential advertising achieve the same media profits as the interactive format? Given the results in Section 3, the answer can be readily obtained as follows (with a graphical representation in Figure 3a).

Proposition 5 (Profitability of Media Formats): *Ad informativeness always weakly increases media profit for all formats; Furthermore,*

1. *For highly informative ads,* $\lambda > \frac{m}{v_c}$*:* $\Pi_{Interactive} = \Pi_{Static} > \Pi_{Sequential}$*,*
2. *For moderately informative ads,* $\frac{m}{v_c+v_a} < \lambda < \frac{m}{v_c}$*:* $\Pi_{Interactive} = \Pi_{Sequential} > \Pi_{Static}$*;*
3. *For less informative ads,* $\lambda < \frac{m}{v_c+v_a}$*, then* $\Pi_{Interactive} = \Pi_{Sequential} = \Pi_{Static}$*.*



(a) X-axis: Ad Informativeness λ (with $m = 1$)

(b) X-axis: Media Content Quality m (with $\lambda = 1$)

Figure 3: Media Profits under Different Formats ($v_c = v_a = 1, r = 0.5$)

[26] For example, in a game of competitive first-degree price discrimination, firms can become worse off if they are able to price discriminate, because competing firms may trigger intensified price competition over a diverse group of consumers. In a dynamic game of durable-good pricing, being able to change prices can trap a monopolist into marginal-cost pricing (i.e., Coase conjecture).

If an ad is sufficiently informative, $\lambda > m/v_c$, then interactive advertising becomes equivalent to static advertising. In this case, the medium benefits more from matching the advertiser and consumer than from profiting solely from the media content. It is optimal for the medium to let the consumer pay attention to the ad until a signal arrives. Both static advertising and interactive advertising achieve exactly that. In contrast, sequential advertising can make the medium worse off due to its restriction on the ad exposure time.

If an ad is moderately informative, $m/(v_c + v_a) < \lambda < m/v_c$, then interactive advertising is reduced to sequential advertising. This result immediately follows from Proposition 4: it is sufficient for the medium to let the consumer choose the media content after showing her the ad for a period of time, because she is motivated to do so anyway. However, under static advertising, granting the consumer complete control of attention encourages her to skip the ad entirely. This approach leaves money on the table because the medium could have extracted the value from matching the advertiser and the consumer.

Last, if an ad is relatively uninformative, $\lambda < m/(v_c+v_a)$, all three advertising formats lead to the same outcome: the medium finds it optimal to stop offering advertising to focus on the media content. Note that the threshold for this to occur increases with higher value of the media content, but decreases with higher match values between the consumer and advertiser. This observation provides a complementary perspective to the role of uninformative advertising as an invitation to search suggested by Mayzlin and Shin [2011]. While an uninformative ad can be employed by a high-quality firm to signal its quality, this comes at the (opportunity) cost of the medium, which could potentially profits more from media content. Hence, to compensate the medium, a sufficiently large v_a is necessary. That is, the condition can be satisfied even for an uninformative ad as long as v_a is sufficiently large.

Overall, as illustrated in Figure 3a, the medium's profit increases with the informativeness of an ad, regardless of the format. Hence, improving the efficiency of an ad is generally beneficial to any media. The same conclusion can be reached with respect to the expected values of an ad match, v_a and v_c. Higher ad values generally improve the profit of a medium of any format.

Figure 3b presents an alternative angle to understand the impact of media format on profits along the dimension of content quality. When the content is unattractive,

$m < \lambda v_c$, the sequential format is suboptimal compared to the interactive and static formats. The performance gap widens as the content quality improves. However, when the media content becomes moderately attractive, $\lambda v_c < m < \lambda(v_c+v_a)$, the consumer is not in the mood for advertising and thus becomes more inclined to skip the ad. The sequential format allows the medium to divert the consumer's attention to the ad. In contrast, the static format loses the consumer's attention entirely, thereby shutting down the ad revenue. Hence, the static medium experiences a drop in the profit. Subsequently as the content quality continues to improve, all three formats lead to increased media profit as the content becomes increasingly more attractive. The above discussion can be summarized as follows.

Corollary 1 (Comparative Statics of Media Profit): *The medium's profit*

1. *weakly increases with ad informativeness λ,*
2. *weakly increases with ad values to consumer and advertisers, v_c and v_a,*
3. *increases with content quality m under the sequential format; but follows a non-monotonic pattern under either the static or interactive format: it first increases until when $m = \lambda v_c$, at which point it drops to a lower level and then resumes increasing with higher m.*

Let us conclude this subsection with three remarks on the implication of Proposition 5 and Corollary 1.

Superiority of digital advertising. Conventional wisdom suggests that it is the ability to target specific consumer segment or individuals that has fueled the growth of digital advertising. Proposition 5 highlights that the *interactive* nature of digital media can also be an important driver. Traditional print and broadcast media lack the technological capability to engage in instantaneous two-way communication, which can help balance between media consumption and ad exposure. The interactive feature can become even more important as traditional media become increasingly equipped with targeting capability.

Completely vs. partially skippable ads. Proposition 5 sheds light on when digital advertising is simply reduced to the more traditional sequential and static formats. This insight is particularly relevant considering that, in practice, skippable ads can be either *completely* skippable (i.e., ads can be skipped from the start) or *partially* skippable (i.e., ads can only be skipped after a minimum run time). Completely skippable ads are in theory equivalent to static advertising with the key feature of allowing consumers to have the full control of their attentions. This approach works well when an ad is sufficiently informative (relative to media content) as both the medium and consumers benefit from the efficiency. In contrast, partially skippable ads are essentially the same as sequential advertising with the feature of limiting the consumers' attention control. This situation arises when an ad is neither too informative nor too uninformative, leading to misaligned incentives between consumers and advertisers.

Media environment for advertising. Part 3 of Corollary 1 underscores an important consideration in ad placement — the media content where the ad is to be inserted. This can be understood in two dimensions. First, media content may vary across different media channels. It is then important for advertisers to decide *where* to advertise. Second, media content can also vary over time, posing a practical question for advertisers, media platforms, or content creators to decide *when* to advertise. Platforms like YouTube, for example, offer various ad placements — pre-roll, mid-roll, and post-roll. Our model can be extended to capture time-varying content, an extension to be explored in Section 5.3.

4.2 Advertising Externality to Consumers

There have been mixed views on how consumers perceive advertising. Advertisements are generally believed to provide product or brand information, assisting consumer decision-making (Nelson 1974) and thus implying a positive benefit to consumers.[27] However, this perspective overlooks the role of advertising medium, which can result in consumer resistance. Common complaints from consumers about advertising include, for example, "the ad is too distracting", "it is just wasting my time", and "it is not relevant/helpful at all". Hence, many analyses on two-sided media markets assume that advertising is generally a nuisance to consumers (e.g., Dukes and Gal-Or 2003, Dukes 2004, Anderson and Coate 2005, Anderson and Gans 2011), with some cautiously acknowledging that this may not hold true for all media formats (e.g., Kaiser and Song 2009). Notably, annoyance from radio and television advertisements is more pronounced, while print media such as newspaper and magazines often elicit a positive perception of ad-

[27] At the very least, consumers incur minimal loss by refraining from purchase if an ad fails to provide substantial information.

vertising.[28]

We provide a unified analysis to reconcile these divergent perspectives on advertising. Such an analysis can illuminate how different media formats affect consumers' attitude towards advertising. The insights can potentially inform multi-format media platforms who factor in consumer reactions in deciding which format to adopt, as well as advertisers who prioritize consumer attitudes in selecting suitable media outlets.

To begin, we first define the concept of advertising externality to a consumer. We note that a frequent complaint in the ad nuisance narratives is that ads divert consumers' attention from media content. Hence, the time lost for content consumption due to advertising can serve as the basis for the notion of advertising externality. Let $V^0 = m/r$ denote the consumer's expected value under a counterfactual scenario, where only the media content is available and no ad is shown. Let V^* denote the total value the consumer expects to gain on the medium by following the optimal attention strategy in equilibrium.[29] Drawing on the literature on media markets and advertising (e.g., Becker and Murphy 1993, Anderson and Gabszewicz 2006, Dukes et al. 2022), we define the externality of advertising, ΔV, as follows:

Definition 2 (Advertising Externality): *The externality of advertising to the consumer is the difference between the expected value of the optimal attention strategy under advertising and the expected value when no advertising is present:* $\Delta V \equiv V^* - V^0$.

We next consider the adverting externality in order. First, it is straightforward that negative externality can hardly arise under static advertising.

Proposition 6 (Externality under Static Advertising): *The externality of static advertising to the consumer is always nonnegative:* $\Delta V_{Static} \geq 0$.

To prove the result, consider first $m > \lambda v_c$, then $V^*_{Static} = V^0 = \frac{m}{r}$. If alternatively $m < \lambda v_c$, then $V^*_{Static} = \frac{\lambda(rv_c+m)}{r(\lambda+r)} > \frac{m}{r}$.

Proposition 6 states that there is no nuisance cost of advertising as long as the consumer adheres to the optimal attention strategy under static advertising. Intuitively,

[28]A survey study conducted by HubSpot in 2016 reveals that people generally dislike digital ads: 73% of the responses dislike online pop-ups, 57% dislike online video ads. (See more detail at https://blog.hubspot.com/marketing/why-people-block-ads-and-what-it-means-for-marketers-and-advertisers.) These findings are in stark contrast with traditional ads: only 36% of responses dislike TV ads and the rate is reduced to 18% to magazine or print ads.

[29]Note that this value will be fully extracted by the medium.

consumers with full control over the choice to engage with either the media content or an ad should not be bothered by the presence of advertising as she can always choose to disregard it.

Turning to sequential advertising, the externality of advertising is measured by comparing $V^*_{Sequential}$ with the no-advertising benchmark $V^0 = m/r$. The following proposition presents the result of this comparison.

Proposition 7 (Externality under Sequential Advertising): *There exists a threshold $\hat{\lambda}$ such that the externality of sequential advertising to the consumer, $\Delta V_{Sequential}$, is positive if $\lambda > \hat{\lambda}$, negative if $m/(v_c + v_a) < \lambda < \hat{\lambda}$, and zero if $\lambda < m/(v_c + v_a)$.*

PROOF: See the appendix. ∎

The key distinction between Proposition 7 and Proposition 6 is that, under sequential advertising, as the medium has more control over what the consumer can attend to, the consumer may be hurt by having advertising. While this conclusion makes intuitive sense, it is not at all immediately evident *a priori*, because the consumer can always choose not to pay attention to the ad. Indeed, under static advertising, the consumer is not hurt precisely because she can withhold her attention at any time.

The situation of sequential advertising is different. The ad operates within a fixed time window, and due to the stochastic arrival of the ad signal, there is always a positive probability that some time is wasted — that is, the consumer receives the ad signal but must await the media content. When an ad is highly informative (i.e., $\lambda > \hat{\lambda}$), the consumer does not loss much if she pays more attention to the ad. However, when the ad is very uninformative (i.e., $\lambda < m/(v_c + v_a)$), it is optimal for the medium to cease advertising. It is only within an intermediate range of ad informativeness that the consumer perceives the ad as a nuisance, as she must wait until S^* to enjoy the media content.

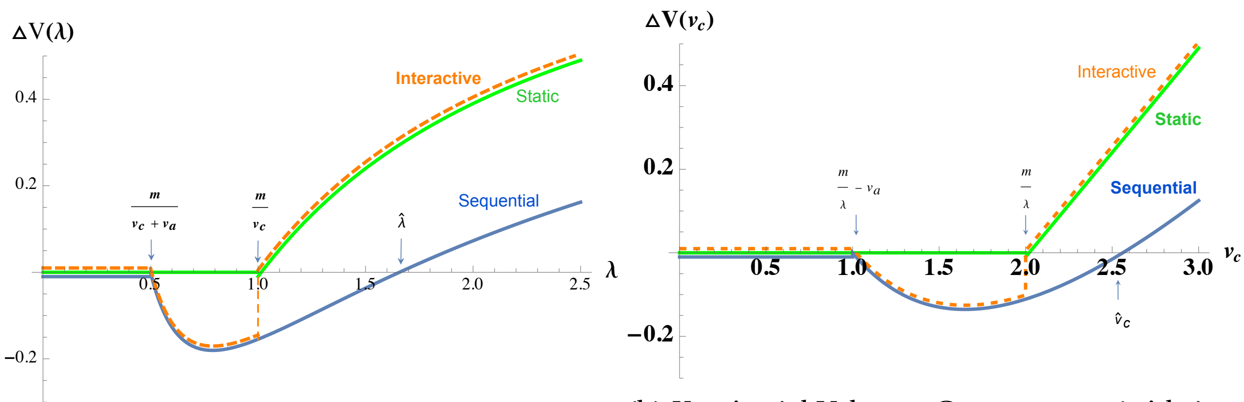
Turning to interactive advertising, we note that, given the equilibrium policy derived in Proposition 4, the consumer's expected value of following the optimal strategy is

$$V^*_{Interactive} = \begin{cases} V^*_{Static} & \text{if } \lambda > m/v_c; \\ V^*_{Sequential} & \text{if } \lambda < m/v_c. \end{cases} \tag{9}$$

It is then straightforward to characterize the advertising externality under this format of advertising. The result is presented in the following proposition:

Proposition 8 (Externality under Interactive Advertising): *The externality of interactive advertising to the consumer,* $\Delta V_{Interactive}$*, is positive if* $\lambda > m/v_c$*, negative if* $m/(v_c + v_a) < \lambda < m/v_c$*, and zero if* $\lambda < m/(v_c + v_a)$*. In addition,* $\Delta V_{Interactive} \geq \Delta V_{Sequential}$*.*

PROOF: See the appendix. ■



(a) X-axis: Ad Informativeness λ (with $v_c = 1$)

(b) X-axis: Ad Value to Consumer v_c (with $\lambda = 0.5$)

Figure 4: Externality of Advertising as a Function of Ad Informativeness ($v_a = 1$, $m = 1$, $r = 0.5$)

Figure 4a compares the patterns of advertising externality across all three advertising formats. Similar to sequential advertising, interactive advertising can also result in a negative externality. As elaborated in Section 4.1, although an interactive ad provides the skip option to the consumer, it may only be *partially* skippable with a minimum exposure time for the ad. As such, consumers may be distracted away from media consumption. Nevertheless, consumers generally expect an overall higher value under interactive advertising than under sequential advertising, *ceteris paribus.* First, the range of parameters for which negative externality occurs is reduced under interactive advertising because $\hat{\lambda} > m/v_c$ (recall that $\hat{\lambda}$ is the threshold that defines the negative externality under sequential advertising). Second, when an ad becomes sufficiently informative, i.e., $\lambda > m/v_c$, interactive advertising is reduced to the static format, enabling consumers to *completely* skip the ad from the start. Thus, consumers become better off than under sequential advertising.

An interesting pattern emerging from the analysis is the non-monotonic impact of ad informativeness on consumer welfare under interactive and sequential advertising.

In practice, one way to improve the informativeness of an ad (i.e., λ) is through increasing the targeting precision. Hence, our results imply that, even for a more targeted ad, consumers may become more annoyed if they cannot easily skip it. This aligns with the empirical finding in Goldfarb and Tucker [2011]: if obtrusive digital ads that are hard to skip are also targeted, they may become less effective. Furthermore, our results suggest that if an ad becomes sufficiently informative, a digital media platform may find it optimal to let consumers decide whether to completely skip it, alleviating the problem of ad annoyance. Conversely, targeting precision might be constrained by privacy protection policies (e.g., removing cookies that can track user browsing behaviors), potentially diminishing the informativeness of an ad. The impact on the externality to consumers will also follow a non-monotonic pattern: it will initially decrease but then increase as the informativeness drops to a certain level.

We should be cautious that λ only captures the informativeness of an ad, influenced in part by a medium's targeting technology. More precise targeting implies a higher ability to match a consumer with a product that delivers higher value, as reflected by v_c. We can derive a similar pattern of the advertising externality with respect to v_c, as illustrated in Figure 4b. For both interactive and sequential advertising, increasing ad value v_c does not always lead to higher ad externality to the consumer. On the contrary, the consumer may experience a stronger nuisance for a higher value of v_c. This is because both formats, to varying degrees, mandate ad exposure to the consumer.

5. Beyond the Simple Model

The preceding analysis has illustrated how the simple framework developed allows us to succinctly analyze the influence of media formats on advertising communication. To make progress, as well as for expositional simplicity, we have made a number of simplifying assumptions. In this section, we briefly discuss a few directions for extending the model to incorporate other aspects of media advertising.

5.1 Multiple Ads

Media often display multiple ads from different advertisers. Our main model can be readily extended to capture this practice. Suppose there are K advertisers available in the advertising market. The k-th advertiser can gain v_{ak} and deliver an expected value of v_{ck} to the consumer if its ad signal reaches her. The ad process of the k-th advertiser follows an exponential process with hazard rate λ_k and is independent across ads. Again, same as the main model, v_{ak}, v_{ck}, and λ_k are common knowledge.

Proposition 9 *Consider a medium offering multiple ads,*

1. *Under the static format, the medium shows the ad from advertiser k as long as $\lambda_k v_{ck} > m$.*
2. *Under the sequential format, the medium shows the ad from advertiser k as long as $\lambda_k(v_{ak} + v_{ck}) > m$, and sequentially in the order of decreasing magnitude of $\lambda_k(v_{ak} + v_{ck})$. Furthermore, the exposure time of each ad is $S_k^* = \frac{1}{\lambda_k} \ln \frac{\lambda_k(v_{ck}+v_{ak})}{m}$.*
3. *Under the interactive format, the medium shows the ad from advertiser k for a minimum period of $S_{mk}^* = \frac{1}{\lambda} \ln \frac{\lambda(v_{ck}+v_{ak})}{m}$ as long as $\lambda v_{ck} < m < \lambda(v_{ck} + v_{ak})$, and $S_{mk}^* = 0$ otherwise. After S_{mk}^*, the skip option is introduced. Furthermore, the ads are shown sequentially in the order of decreasing magnitude of $\lambda_k(v_{ak} + v_{ck})$.*

Proposition 9 indicates that the decision to show an ad depends on a threshold rule that is independent across ads. This is quite natural given the independence assumption. Importantly, the proposition provides insights into how media should prioritize different ads. In the case of static and sequential formats, ads with higher index values $\lambda_k(v_{ak}+v_{ck})$ should be prioritized and displayed first. For advertisers planning marketing campaigns on media outlets, understanding the positioning of their ads in prospective media may be useful. A lower value of $\lambda_k(v_{ak} + v_{ck})$ will put them at a disadvantage. To enhance their position, advertisers can make ads more informative, such as providing substantial and clear information about the brand or product to facilitate consumers' ad processing. Improving the total ad value $v_{ak} + v_{ck}$ is another avenue to secure a better position.[30] Therefore, although the model does not explicitly model competition among advertisers, it suggests that advertisers should compete for a higher ranking position on media by enhancing the value of $\lambda_k(v_{ak} + v_{ck})$.

The third part of Proposition 9 also illustrates that when a medium has the interactive capabilities, some of its ads are partially skippable whereas others are completely

[30] To some extent, the total ad value (i.e., total surplus) is constrained by the nature of the product. If the advertiser provides more value to the consumer, then it implies forgoing some surplus. This redistribution of surplus does not affect the preference ranking by the media in our model.

skippable. This pattern aligns with some leading online video platforms, such as YouTube, which offers different ad formats.

5.2 Ads with Cognitive Cost and Entertainment Value

In our simple model, paying attention to either the media content or the ad does not involve any cognitive effort of the consumer. This assumption is made without loss of generality if the cognitive cost of attending to the ad is comparable to that of attending to the media content. However, in reality, consumers often exert more effort in processing an ad than paying attention to the media content. For example, when a consumer is relatively new to the product category of an advertised product, she may need to think harder whether the product is a good fit to her or not.

Furthermore, the baseline model assumes the value of an ad materializes only upon the arrival of ad signal. However, in reality, ads often deliver more than just product information. To capture attention, advertisers deliberately design ads to be interesting, humorous, and entertaining, sometimes incorporating storytelling elements (Dukes and Liu 2020). Such ads typically offer benefits to consumers beyond conveying product information.

To account for the differential cognitive costs and entertainment value associated with an ad, let us continue to normalize the cognitive cost of paying attention to the media content to be zero, but allow the ad attention process to incur a flow cost — cognitive effort c (with a slight abuse of notation), and a flow benefit — entertainment value w, independent of the ad's signal arrival process. To examine the implications of this additional feature, let us first consider the case of static advertising. We can define a Gittins index in a similar manner by extending Equation 1:

$$G_A(x) \equiv \sup_{t>x} \frac{\int_x^t [v_c e^{-rs} + \int_x^s (w-c)e^{-rk}dk] f(s)ds}{\int_x^t [1-F(s)]e^{-rs}ds}, \tag{10}$$

where in the numerator, $v_c e^{-rs}$ is the discounted expected reward upon signal arrival, and $\int_x^s (w-c)e^{-rk}dk$ is the cumulative reward (net of cognitive cost) up to the point of signal arrival. Due to the memoryless feature of exponential distribution $F(\cdot)$, the index does not depend on the process time x (i.e., conditional on the event that the signal has not arrived, the likelihood of arrival still follows the same distribution). Then this index value continues to satisfy the monotonicity condition — with probability one, the index value is non-increasing (Gittins et al. [2011]). As such, the index value reduces to the per-period reward, which in this case is $\lambda v_c + w - c$.

Proposition 10 *Under static advertising, the consumer pays attention to the ad if and only if $\lambda v_c + w - c > m$. The medium advertises as long as $\lambda v_c + w - c > m$.*

Under sequential advertising, the consumer's decision rule remains the same as we can continue to apply the finite-horizon equivalence result here (Weber 1992). However, the medium's optimal decision becomes less tractable because the medium now needs to take into account the utility gain (or loss) due to cognitive cost and entertainment value. Specifically, Equation 5 now becomes

$$\max_{S\geq 0} \Pi_{Sequential}(S) = \int_0^S \left[(v_c + v_a)e^{-rs} + \int_0^s (w-c)e^{-rk}dk\right]dF(s) + \int_S^\infty me^{-rs}ds.$$

The solution S^* can be found by the first-order condition

$$\lambda(v_c + v_a)e^{-(\lambda+r)S^*} + \frac{\lambda(w-c)}{r}(1-e^{-rS^*})e^{-\lambda S^*} - me^{-rS^*} = 0. \tag{11}$$

Proposition 11 *Under sequential advertising, the consumer pays attention to the ad as long as $\lambda v_c + w - c > b$ until the signal has arrived or the ad exposure has ended, whichever comes first. The medium shows an ad with constant entertainment value for a duration of S^* determined by Equation 11. Furthermore, S^* increases with w but decreases with c.*

The proof of the first two parts of Proposition 11 follows from the main analysis straightforwardly. The third result related to comparative statics with respect to cognitive cost and entertainment value applies the monotone comparative statics. The proposition says that if the consumer incurs more cognitive cost to process the ad, the medium will also want to shorten the ad exposure time to reduce the negative impact on consumer utility.

The proposition also illustrates that as an ad contains more entertainment value, consumers become more willing to process the ad. Furthermore, the medium becomes more willing to extend the ad exposure time, which in turn increases the chance that consumers will receive the ad signal. The complementarity between entertainment value and informational value to some extent explains why a considerable fraction of television commercials is devoted to entertaining consumers.

5.3 Time-Varying Content Value

Another direction for enriching the model is to allow time-varying media content. To illustrate the general idea, let us focus on the case of sequential advertising.[31] A few additional assumptions and notations are necessary. Assume that the content process follows a Markov process captured by state $y(t)$, which is independent of the advertising process. Whenever the consumer processes the content at time t, she obtains a random reward $m(y(t))$. When the medium switches from the content to the ad, the consumer freezes the content process but activates the ad process, with the content state $y(t)$ remaining unchanged.

Proposition 12 *Under time-varying media content, a medium shows an ad whenever $\lambda(v_a + v_c) > M^*(y)$, where $M^*(y)$ is defined as the solution to the following equation:*

$$M^*(y) \equiv \sup_{\tau} \frac{\mathbb{E}[\int_0^{\tau} m(y(s))e^{-rs}ds|y(0) = y]}{\mathbb{E}[\int_0^{\tau} e^{-rs}ds|y(0) = y]}. \tag{12}$$

This result follows straightforwardly from the index policy. The ad allocation policy in the main analysis can be understood as a special case of the above result. In the more general media environment in which content value fluctuates, the optimal policy in Proposition 12 implies that a medium may insert an ad a number of times throughout a content exposure session. This is consistent with the observation that television programs place commercials throughout the broadcasting period.

6. Concluding Remarks

Despite the reliance of advertising on media to reach consumers, the influence of media formats on advertising communication has received relatively little attention in the advertising literature. We present a unifying framework that allows us to understand the fundamental differences and equivalences among various media formats. The model addresses the attention-steering problem of media, which must allocate ads and media content to direct consumers' attention over time. Based on media's capability to steer consumer attention, we can classify advertising formats into three basic types: static,

[31] The model here remains silent on many institutional details that may further influence media behavior. Zhou [2004] provides an early analysis on how a monopoly television network can structure commercial breaks based on the appeal of a program.

sequential, and interactive. We show how these different formats grapple with two fundamental problems in media advertising: incentive misalignment between consumers and media, and the media's inability to observe whether consumers have received product match signals. The analysis sheds light on several issues such as how digital advertising fundamentally differs from traditional advertising, how they evolve in light of the improving (and regulated) targeting technology, and what causes the consumer heterogeneity in ad attitudes on different media.

Clearly, this preliminary study raises more questions than it answers. Nevertheless, as shown in Section 5, the model can be flexibly extended in several ways. Still, additional avenues warrant exploration. The simple model presented here neglects consumer heterogeneity. An obvious area of inquiry is to examine the implications of consumer heterogeneity for advertising allocation policy, particularly in cases where media lack the capability of targeting individuals or the targeting technologies are imperfect. As consumer heterogeneity is assumed away, pricing decisions are also nullified in this study, allowing us to focus on the information design problem. Future research could examine the interplay between pricing and information design. Another interesting avenue for future studies is how multiple media may jointly affect advertising communication. This is particularly relevant when media platforms offer multiple channels or when multiple media platforms are competing for consumer attention. Additionally, we have made a small step in exploring the time dimension in media formats of advertising by assuming a simple and tractable information structure. This simplification to some extent has restricted the media formats considered. In practice, more complex information structures may lead to a larger strategy space for a medium and thus more intricate media formats. For example, considering multiple ad signals, fatigue, or interactions between media content and ads introduces a broader set of decisions for a medium. Exploration along this line might be fruitful. Lastly, although our model assumes a fully rational consumer with forward-looking capability, the optimal attention strategy is surprisingly myopic (or involving one-step look-ahead). Future studies dealing with complex and intractable media environments might find it beneficial to consider myopic consumers to simplify analysis without much loss of generality.

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Appendix

A. Proof of Lemma 4

To prove this lemma, we first prove a set of useful intermediate results.

Lemma A.1 (Advantage of Optional Control (Part I)): *Suppose the ad signal has not arrived at time t_0, and consider an exogenous deadline T (possibly infinite). Then within the period $[t_0, T]$, running the media content exclusively is weakly dominated by offering and committing to the optional control $\{A, M\}$ throughout.*

PROOF: First, consider the case $\lambda v_c \leq m$. By Lemma 3, the consumer will always choose to consume the media content M when the optional control $\{A, M\}$ is available. Hence, the total profit the media can extract under the optional control is the same as that under running M exclusively.

Second, consider the case $\lambda v_c > m$. By Lemma 3, when the optional control $\{A, M\}$ is available within $[t_0, T]$, the consumer will first pay attention to the ad A until the signal arrival, or the deadline T, whichever comes first. There are two possible outcomes, depending on the arrival of the ad signal.

Case (a): The ad signal does not arrive before T (i.e., $\sigma > T$). In this case, the consumer pays attention to the ad throughout $[t_0, T]$. The profit accrued is then higher than the profit under running M exclusively because $\lambda v_c > m$.

Case (b): The ad signal eventually arrives before T (i.e., $\sigma < T$). In this case, the profit yielded after σ is the same across the two policies because both lead the consumer to consume M only. However, within $[t_0, \sigma]$, the optional-control policy allows the consumer to process the ad. Thus, by the same argument in Case (a), the profit is higher under the optional-control policy.

Summarizing, by committing to the optional control within $[t_0, T]$, the medium can achieve a higher profit than running the media content exclusively throughout. ∎

Lemma A.2 (Advantage of Optional Control (Part II)): *Suppose the ad signal has not arrived at time t_0, and consider an exogenous deadline T (possibly infinite). Suppose further that $\lambda v_c > m$. Then within the period $[t_0, T]$, exposing the ad exclusively for a period of S followed by running the media content exclusively is weakly dominated by offering and committing to the optional control $\{A, M\}$ throughout.*

PROOF: When $\lambda v_c > m$, by Lemma 3, under the optional control $\{A, M\}$ the consumer will first pay attention to the ad A until the signal arrival, or the deadline T, whichever comes first. Let us examine the two possible cases of signal arrival.

Case (a): The ad signal does not arrive before S (i.e., $\sigma > S$). In this case, prior to $t = S$, the consumer pays attention to the ad under both policies. After $t = S$, however, the consumer continues to process the ad until the signal arrival under the optional-control policy, whereas she will be diverted to media content under the sequential policy. Hence, starting from $t = S$, the situation is the same as that in Lemma A.1. Then, the optional-control policy dominates the sequential policy.

Case (b): The ad signal arrives before S (i.e., $\sigma < S$). In this case, prior to $t = \sigma$, the consumer pays attention to the ad under both policies. After $t = S$, however, the consumer switches to the media content under the optional-control policy, whereas she has to turn to the outside option for the period of $[\sigma, S]$ under the sequential policy. Thus, there is profit gain under the optional-control policy.

Summing up, by committing to the optional control within $[t_0, T]$, the medium can achieve a higher profit than running the ad and media content sequentially throughout.

■

We are now prepared to prove Lemma 4.

PROOF: By Lemma A.1 and Lemma A.2, the optimal media policy entails sequencing of two components: running the ad A exclusively and running the optional control $\{A, M\}$. This is because any M-exclusive run will be weakly dominated by the optional control $\{A, M\}$. Under the optional control $\{A, M\}$, the medium may or may not commit the course of action throughout. Next, we show that the medium has no incentive to overturn the consumer's decision whenever the optional control $\{A, M\}$ is introduced. Take the first sequence such that the medium starts with running A exclusively for a period of S_1, followed by a period S_2 of optional control $\{A, M\}$.

Case (1): $\lambda v_c \leq m$. By Lemma 3, the consumer will keep paying attention to the ad during the period of $[0, S_1]$ unless the ad signal has arrived. If the ad signal does arrive before S_1, that is, $\sigma < S_1$, then it will be always optimal for the consumer to choose M after $t = S_1$, provided it is available. If at any point later than S_1, the medium disables the access to M, then it will incur a loss of profit. If the ad signal does not arrive before S_1, that is, $\sigma > S_1$, then the consumer does not have any incentive to process A after

$t = S_1$. Given the availability of M after $t = S_1$, the consumer will consume it. Therefore, regardless of whether the signal has arrived or not, the consumer will choose M as long as it is available. Hence, by observing the decision of the consumer at the point of introducing $\{A, M\}$, the medium does not update its information thereafter. Its optimal policy is to let the consumer continue to consume the media content M. This can be achieved by either continuing to offer the optional control $\{A, M\}$ or restricting to the media content M.

Case (2): $\lambda v_c > m$. Again, by Lemma 3, the consumer will keep paying attention to the ad during the period of $[0, S_1]$ unless the ad signal has arrived. If the ad signal does arrive before S_1, that is, $\sigma < S_1$, then it will be always optimal for the consumer to choose M after $t = S_1$, provided it is available. However, if the ad signal does not arrive before S_1, that is, $\sigma > S_1$, then the consumer will continue to process the ad after $t = S_1$. If the medium wants to overturn the consumer's decision, then it can instead introduce the media content exclusively, implying the failure to commit to the optional control $\{A, M\}$ throughout $[S_1, S_2]$. However, according to Lemma A.2, this is weakly dominated by the committed optional control. Therefore, it is never optimal for the medium to violet the commitment for any point after $t = S_1$.

The two cases taken together imply that once the medium has introduced the optional control $\{A, M\}$, it is committed to providing this option thereafter.

■

B. Proof of Proposition 7

The proof evokes a variant of Bernoulli's inequality, which is stated below without proof:

Lemma B.1 (A Variant of Bernoulli's Inequality): $(1 + y)^\theta < \frac{1+y}{1+y-\theta y}$*, for any real numbers* $\theta > 1$ *and* $y \in (-1, 0)$.

We next present the proof of Proposition 7.

PROOF: The total discounted value a consumer expects following the optimal strategy is:

$$V^*_{Sequential} = \int_0^{S^*} v_c e^{-rt} dF(t) + \int_{S^*}^{\infty} m e^{-rt} dt \tag{A-1}$$

$$
\begin{aligned}
&= \frac{\lambda v_c}{r+\lambda}(1-e^{-(r+\lambda)S^*}) + \frac{m}{r}e^{-rS^*} && \text{(A-2)}\\
&= \frac{\lambda v_c}{r+\lambda}(1-e^{-(r+\lambda)S^*}) + \frac{\lambda(v_c+v_a)}{r}e^{-(r+\lambda)S^*} && \text{(A-3)}\\
&= \frac{\lambda v_c}{r+\lambda} + \big(\frac{\lambda(v_c+v_a)}{r} - \frac{\lambda v_c}{r+\lambda}\big)e^{-(r+\lambda)S^*} && \text{(A-4)}\\
&= \frac{\lambda v_c}{r+\lambda} + \big(\frac{\lambda(v_c+v_a)}{r} - \frac{\lambda v_c}{r+\lambda}\big)\big(\frac{m}{\lambda(v_c+v_a)}\big)^{1+\frac{r}{\lambda}}. && \text{(A-5)}
\end{aligned}
$$

where the third equality (A-3) follows from the first-order condition in Equation 6, and the last equality follows from the solution in Equation 7. Let $\Delta V_{Sequential} = V^*_{Sequential} - V^0$, where $V^0 = m/r$. Note that, if $\lambda < m/(v_c+v_a)$, sequential advertising entails zero ad exposure, and thus the externality is zero, $\Delta V_{Sequential} = 0$. We only need to consider what happens if $\lambda > m/(v_c+v_a)$. Let us analyze two cases.

Case 1: $m/(v_c+v_a) < \lambda < m/v_c$. We shall show that $\Delta V_{Sequential} < 0$ under this case. Note that this case is equivalent to $\lambda v_c < m < \lambda(v_a+v_c)$. It would be more convenient to characterize the change of $\Delta V_{Sequential}$ with respect to m. Note that

$$
\frac{\partial \Delta V_{Sequential}}{\partial m} = \underbrace{\big(\frac{\lambda(v_c+v_a)}{r} - \frac{\lambda v_c}{r+\lambda}\big)}_{>0}\big(\lambda(v_c+v_a)\big)^{-1-\frac{r}{\lambda}}\big(1+\frac{r}{\lambda}\big)m^{\frac{r}{\lambda}} - \frac{1}{r} < 0
$$

for m sufficiently small, and

$$
\frac{\partial^2 \Delta V_{Sequential}}{\partial m^2} = \big(\frac{\lambda(v_c+v_a)}{r} - \frac{\lambda v_c}{r+\lambda}\big)\big(\lambda(v_c+v_a)\big)^{-1-\frac{r}{\lambda}}\big(1+\frac{r}{\lambda}\big)\frac{r}{\lambda}m^{\frac{r}{\lambda}-1} > 0.
$$

Together with the facts that $\Delta V_{Sequential}(m=0) = \lambda v_c/(r+\lambda)$ and that $\Delta V_{Sequential}(m = \lambda(v_c+v_a)) = 0$, it follows that, there exists a threshold $\hat{m} \in (0, \lambda(v_c+v_a))$ such that $\Delta V_{Sequential} > 0$ if $m < \hat{m}$, and $\Delta V_{Sequential} < 0$ if $m \in (\hat{m}, \lambda(v_a+v_c))$. Furthermore, we need to show that $\hat{m} < \lambda v_c$ to complete the proof of this case. It suffices to prove that if $m = \lambda v_c$, $\Delta V_{Sequential} < 0$. Indeed, following Equation A-5 we have

$$
\begin{aligned}
\Delta V_{Sequential} \equiv V^*_{Sequential} - V^0 &= \frac{\lambda v_c}{r+\lambda} + \big(\frac{\lambda(v_c+v_a)}{r} - \frac{\lambda v_c}{r+\lambda}\big)\big(\frac{v_c}{v_c+v_a}\big)^{1+\frac{r}{\lambda}} - \frac{\lambda v_c}{r}\\
&= -\frac{\lambda^2 v_c}{r(r+\lambda)} + \frac{\lambda\big(\lambda(v_c+v_a)+rv_a\big)}{r(r+\lambda)}\big(\frac{v_c}{v_c+v_a}\big)^{1+\frac{r}{\lambda}},\\
&< -\frac{\lambda^2 v_c}{r(r+\lambda)} + \frac{\lambda\big(\lambda(v_c+v_a)+rv_a\big)}{r(r+\lambda)} \cdot \frac{\lambda v_c}{\lambda(v_c+v_a)+rv_a}
\end{aligned}
$$

$$= 0,$$

where the second equality is obtained by rearranging terms and the inequality follows from Lemma B.1 by noticing that $y = -v_a/(v_a + v_c)$ and $\theta = 1 + r/\lambda$.

Case 2: $\lambda > m/v_c$. We shall show that under this case, there exists $\hat{\lambda}$ such that, $\Delta V_{Sequential} < 0$ if $\lambda \in (m/v_c, \hat{\lambda})$, but $\Delta V_{Sequential} > 0$ if $\lambda > \hat{\lambda}$. This can be proved by demonstrating that (a) $\Delta V_{Sequential} < 0$ at $\lambda = m/v_c$, (b) $\Delta V_{Sequential} > 0$ at $\lambda \to \infty$, and (c) $\partial \Delta V_{Sequential}/\partial \lambda > 0$. Note that (a) has been proved above already. (b) is easily checked by taking $\lambda \to \infty$ under Equation A-5. The last step is to verify (c). Note that

$$\frac{\partial \Delta V_{Sequential}}{\partial \lambda} = \frac{(\frac{\lambda(v_c+v_a)}{m})^{-1-\frac{r}{\lambda}}}{\lambda(1+\lambda)^2} \cdot H$$

where

$$\begin{aligned} H =& v_c r \lambda (\frac{\lambda(v_c + v_a)}{m})^{1+\frac{r}{\lambda}} - \big(v_a r^2 + (v_c + 2v_a) r\lambda + v_a \lambda^2\big) \\ &+ (r+\lambda)(v_a r + (v_a + v_c)\lambda) \ln \frac{\lambda(v_c + v_a)}{m}. \end{aligned}$$

It suffices to show that $H > 0$ for $\lambda > m/v_c$, or equivalently, $m < \lambda v_c$. Note that $\partial H/\partial m < 0$. The only thing left to show is $H(m = \lambda v_c) > 0$. Indeed,

$$\begin{aligned} H|_{m=\lambda v_c} =& v_c r \lambda (\frac{v_c + v_a}{v_c})^{1+\frac{r}{\lambda}} - \big(v_a r^2 + (v_c + 2v_a) r\lambda + v_a \lambda^2\big) \\ &+ (r+\lambda)(v_a r + (v_a + v_c)\lambda) \ln \frac{(v_c + v_a)}{v_c} \\ >& v_c r \lambda (\frac{v_c + v_a + \frac{r}{\lambda} v_a}{v_c}) - \big(v_a r^2 + (v_c + 2v_a) r\lambda + v_a \lambda^2\big) \\ &+ (r+\lambda)(v_a r + (v_a + v_c)\lambda) \ln \frac{(v_c + v_a)}{v_c} \\ >& v_c r \lambda (\frac{v_c + v_a + \frac{r}{\lambda} v_a}{v_c}) - \big(v_a r^2 + (v_c + 2v_a) r\lambda + v_a \lambda^2\big) \\ &+ (r+\lambda)(v_a r + (v_a + v_c)\lambda) \frac{v_a}{v_c + v_a} \\ =& \frac{(r+\lambda) r v_a^2}{v_a + v_c} \\ >& 0. \end{aligned}$$

The first inequality makes use of the Bernoulli's Inequality in Lemma B.1 by taking again

$y = -v_a/(v_a + v_c)$ and $\theta = 1 + r/\lambda$. The second inequality follows from the standard logarithm inequality, i.e., $\ln(1 + z) \geq z/(1 + z)$ for $z > -1$.

■

C. Proof of Proposition 8

PROOF: Part(i): Based on Equation 9, we have

$$\Delta V_{Interactive} \equiv V^*_{Interactive} - V^0 = \begin{cases} V^*_{Static} - V^0 & \text{if } m < \lambda v_c; \\ V^*_{Sequential} - V^0 & \text{if } m > \lambda v_c. \end{cases}$$

Then both Proposition 6 and Proposition 7 immediately suggest that (a) $\Delta V_{Interactive} = \Delta V_{Sequential} = \Delta V_{Static} = 0$ if $\lambda < m/(v_c + v_a)$, (b) $\Delta V_{Interactive} = \Delta V_{Sequential} < 0$ if $m/(v_c + v_a) < \lambda < m/v_c$, and (c) $\Delta V_{Interactive} = \Delta V_{Static} > 0$ if $\lambda > m/v_c$.

Part(ii): To prove the second result that $\Delta V_{Interactive} \geq \Delta V_{Sequential}$, it suffices to show $V^*_{Static} > V^*_{Sequential}$ if $m < \lambda v_c$. Indeed,

$$\begin{aligned}
V^*_{Static} - V^*_{Sequential} &= \frac{\lambda(rv_c + m)}{r(\lambda + r)} - \Big[\frac{\lambda v_c}{r + \lambda} + \big(\frac{\lambda(v_c + v_a)}{r} - \frac{\lambda v_c}{r + \lambda}\big)\big(\frac{m}{\lambda(v_c + v_a)}\big)^{1+\frac{r}{\lambda}}\Big] \\
&= \frac{\lambda}{r(r + \lambda)}\Big[m - \big(\lambda(v_c + v_a) + rv_a\big) \cdot \big(\frac{m}{\lambda(v_c + v_a)}\big)^{1+\frac{r}{\lambda}}\Big] \\
&> \frac{\lambda}{r(r + \lambda)}\Big[m - \big(\lambda(v_c + v_a) + rv_a\big) \cdot \frac{m}{(r + \lambda)(v_c + v_a) - \frac{rm}{\lambda}}\Big] \\
&= \frac{m\lambda}{r(r + \lambda)}\Big[1 - \frac{\lambda(v_c + v_a) + rv_a}{(r + \lambda)(v_c + v_a) - \frac{rm}{\lambda}}\Big] \\
&> \frac{m\lambda}{r(r + \lambda)}\Big[1 - \frac{\lambda(v_c + v_a) + rv_a}{(r + \lambda)(v_c + v_a) - rv_c}\Big] \\
&= 0,
\end{aligned}$$

where the first inequality makes use of Lemma B.1 by treating $y = m/\lambda(v_a + v_c) - 1$ and $\theta = 1 + r/\lambda$, and the second inequality follows from $m < \lambda v_c$. ■